AQA 3.1.4 and 3.1.10 Equilibrium constants K_c and K_p

Answer in AQA exam style: use the marks as a guide to how much to write and show all working in calculations. Mark your work with the mark scheme at the back. Questions marked as practice questions in the mark scheme were written in AQA style; all others are from AQA past papers.

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22 marks · about 25 minutes

Key information

- Only temperature changes the value of K_p or K_c; pressure, concentration and catalysts do not
- Exothermic forward reaction: raising temperature lowers K; endothermic: raising temperature increases K
- Mole fraction = moles of that gas ÷ total moles of gas
- Partial pressure = mole fraction × total pressure
- K_p uses partial pressures in round brackets, e.g. $K_p = p(C)^2 \div p(A)\ p(B)^3$; never square brackets

Section A: Short-answer questions

01 Explain why changing the total pressure has no effect on the position of the equilibrium $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$.

[1 mark]

02 1.00 mol of N_2O_4 is heated and 30% dissociates: $N_2O_4(g) \rightleftharpoons 2NO_2(g)$. Calculate the amount of each gas and the total amount of gas at equilibrium.

[2 marks]

03 When one mole of ammonia is heated to a given temperature, 50% of it dissociates and the following equilibrium is established.

$NH_3(g) \rightleftharpoons ½N_2(g) + 1½H_2(g)$

What is the total amount, in moles, of gas in this equilibrium mixture?

[1 mark]

A 1.5
B 2.0
C 2.5
D 3.0

04 In an equilibrium mixture at a total pressure of 200 kPa, the mole fraction of oxygen is 0.25. Calculate the partial pressure of oxygen.

[1 mark]

Section B: Longer questions

05 $2P(aq) + Q(aq) \rightleftharpoons R(aq) + 3S(aq)$

A 25.0 cm^3 sample of a solution of P is added to a 20.0 cm^3 sample of a solution of Q. The mixture is allowed to reach equilibrium. The amounts in the equilibrium mixture are

P = 0.0145 mol Q = 0.0275 mol R = 0.0115 mol S = 0.0345 mol

05.1 Calculate the amount, in moles, of P before the reaction with Q. Use your answer to calculate the concentration, in mol dm^{-3}, of P in the initial 25.0 cm^3 sample.

[2 marks]

05.2 Give the expression for the equilibrium constant, K_c. Calculate the value of K_c and deduce its units.

[4 marks]

05.3 Explain why the amount of S increases when water is added to the equilibrium mixture.

[2 marks]

06 Nitrogen and hydrogen were mixed in a 1:3 mole ratio and left to reach equilibrium in a flask at a temperature of 550 K.

$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

06.1 When equilibrium was reached, the total pressure in the flask was 150 kPa and the mole fraction of $NH_3(g)$ in the mixture was 0.80. Calculate the partial pressure of each gas in this equilibrium mixture.

[3 marks]

06.2 Give an expression for the equilibrium constant (K_p) for this reaction.

[1 mark]

06.3 In a different equilibrium mixture, under different conditions, the partial pressures of the gases are shown in the table.

Gas	Partial pressure / kPa
N_2	1.20×10^2
H_2	1.50×10^2
NH_3	1.10×10^3

Calculate the value of the equilibrium constant (K_p) for this reaction and give its units.

[2 marks]

06.4 The enthalpy change for the reaction is −92 kJ mol^{-1}. State the effect, if any, of an increase in temperature on the value of K_p for this reaction. Justify your answer.

[3 marks]

Mark Scheme

M1, M2... are separate marking points. / separates acceptable alternatives. Words in brackets are not needed for the mark. ECF = error carried forward. Questions and mark schemes © AQA, reproduced for internal school use.

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Question	Marking guidance	Comments	Mark
01	Equal numbers of moles of gas on each side	*Source: Practice question (written for this set)*	1
02	N_2O_4 0.70 mol **and** NO_2 0.60 mol total 1.30 mol	*Source: Practice question (written for this set)*	2
03	**A**	*Source: 7404/2 Jun 2019 Q12*	1
04	50 kPa	*Source: Practice question (written for this set)*	1
05.1	M1 n(P) = 0.0145 + (2 × 0.0115) = 0.0375 M2 [P] = 0.0375 ÷ 0.025 = 1.50 mol dm^{-3}	*Source: 7405/2 Jun 2024 Q02*	2
05.2	M1 $K_c = [R][S]^3 \div [P]^2[Q]$ M2 inserts values and divides by 0.045 dm^3 M3 = 1.81 to 1.82 M4 mol dm^{-3}	If no use of volume lose M2 but can score M3 for 0.0817	4
05.3	M1 equilibrium shifts to the side with most moles M2 to oppose the decrease in concentration of all species / dilution	Alternative: K_c = (amounts) × 1/V, so if V increases amounts of R and S must increase to keep K_c constant	2
06.1	$p(NH_3) = 0.8 \times 150 = 120$ kPa $p(H_2) = 0.75 \times 30 = 22.5$ or 23 kPa $p(N_2) = 0.25 \times 30 = 7.5$ kPa	*Source: 7405/1 Jun 2018 Q02* Alternative: mole fractions 0.15 and 0.05 × 150	3
06.2	$K_p = p(NH_3)^2 \div (p(N_2) \times p(H_2)^3)$	Penalise [ ]	1
06.3	$K_p = (1.10 \times 10^3)^2 \div ((1.50 \times 10^2)^3 \times 1.20 \times 10^2) = 0.0029$ to 0.0030 kPa^{-2}	Allow 2.9 to 3.0 × 10^{-9} Pa^{-2} If expression inverted, allow 1 mark for kPa^2	2
06.4	M1 decreases M2 equilibrium shifts in the endothermic direction (to the left) M3 to oppose the increase in temperature	If increase or no change, 0 marks	3